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INSPIRING INNOVATION

भौतिक विज्ञान (प्रश्न-पत्र I)

PHYSICS (Paper I)

निर्धारित समय: 2 घंटे 44 मिनट
Time Allowed: 2 Hrs. 44 Min.

Test Code : 5687

अधिकतम अंक: 200
Maximum Marks: 200

QUESTION PAPER SPECIFIC INSTRUCTIONS

(Please read each of the following instructions carefully before attempting questions)

There are **FIVE** questions in all, printed in **ENGLISH**. Each question carries 50 marks.

Question No. 1 is **COMPULSORY**. Of the remaining four questions (Q2 to Q5), any **THREE** are to be attempted. Thus a candidate has to attempt **FOUR** questions in all, totalling 200 marks.

The number of marks carried by a question / part is indicated against it.

Assume suitable data if considered necessary, and indicate the same clearly.

Unless otherwise indicated, symbols and notations carry their usual standard meanings. Vectors are indicated in bold type or by an over-arrow.

Neat sketches and diagrams are to be drawn wherever necessary, and must be labelled.

Attempts of questions shall be counted in sequential order. Unless struck off, attempt of a question shall be counted even if attempted partly. Any page or portion of the page left blank must be clearly struck off.

Physical Constants :

Velocity of light	$c = 3 \times 10^8 \text{ m/s}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J/K}$
Planck constant	$h = 6.627 \times 10^{-34} \text{ J s}$
Permittivity of free space	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
Charge of the electron	$e = 1.6 \times 10^{-19} \text{ C}$
Rest mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
Universal gas constant	$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$

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- 1.(a) A frame S' rotates with a constant angular velocity $\omega = 2 \hat{k} \text{ rad s}^{-1}$ relative to a fixed frame S having the same origin. At a certain instant a particle has position vector $\mathbf{r} = (\hat{i} + 5 \hat{j}) \text{ m}$ and velocity $\mathbf{v}' = (6 \hat{i} - 3 \hat{j}) \text{ m s}^{-1}$ as measured in S' . Determine the Coriolis acceleration and the centrifugal acceleration of the particle at this instant. 10
- 1.(b) A narrow tunnel is bored along a diameter of the Earth. Show that a body dropped into it executes simple harmonic motion, and find its time period. 10
- 1.(c) Find the position of the centre of mass of a uniform semicircular wire (a half-ring) of radius R . 10
- 1.(d) A rod of proper length 1.0 m lies in the $x'y'$ -plane of its rest frame S' , making an angle of 45° with the x' -axis. The frame S' moves with a velocity $0.6c$ along the x -axis relative to the laboratory frame S . Determine (i) the length of the rod and (ii) the angle it makes with the x -axis, as measured in S . 10
- 1.(e) An electron is accelerated from rest until it acquires a kinetic energy of 1.5 MeV . Calculate its speed and its momentum. 10
- 2.(a) A particle moves under the action of an attractive central force in a plane orbit described by $r = a(1 + \cos\theta)$, where a is a constant. Show that the magnitude of the force varies as the inverse fourth power of the distance from the centre of force. Also, calculate the potential energy and the total energy. 20
- 2.(b) A uniform circular disc of mass M and radius R has a concentric circular hole of radius $R/2$ cut out of it. Determine the moment of inertia of the remaining body
- (i) about an axis through its centre perpendicular to its plane, and
- (ii) about a diameter. 15
- 2.(c) Starting from the expression for relativistic momentum $p = \gamma m_0 v$, use the work energy theorem to derive the relativistic kinetic energy $K = (\gamma - 1)m_0 c^2$. Hence show that for $v \ll c$ it reduces to $\frac{1}{2}m_0 v^2$, and that for a particle of total energy E and momentum p the speed is $v = pc^2/E$. 15
- 3.(a) (i) A particle of mass m_1 moving with velocity u collides head-on and elastically with a particle of mass m_2 initially at rest. Derive expressions for the velocities of the two particles after the collision. 10
- (ii) A body of mass 2 kg moving at 6 m s^{-1} collides head-on and elastically with a stationary body of mass 4 kg . Find the velocities of both bodies after the collision. 5
- 3.(b) (i) State Bernoulli's theorem and derive it for the streamline flow of an ideal, incompressible, non-viscous fluid. 10
- (ii) Water flows steadily through a horizontal pipe whose cross-sectional area narrows from 10 cm^2 to 5 cm^2 . If the speed of water in the wider section is 2 m s^{-1} , find the speed in the narrower section and the pressure difference between the two sections. (Density of water = 1000 kg m^{-3} .) 10



- 3.(c) A solid sphere of radius R has a density that varies with the distance r from its centre as $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant. Obtain expressions for the gravitational field due to the sphere at its surface and at a point distant $R/2$ from the centre. 15
- 4.(a) A uniform solid cube of mass M and side a has one of its corners at the origin O , with its three edges directed along the x -, y - and z -axes. Obtain the moment-of-inertia tensor of the cube about the origin O . 15
- 4.(b) (i) State the conditions for a force field to be conservative, and show that such a field can be expressed as the gradient of a scalar potential. 5
- (ii) Show that the force field $\mathbf{F} = (2xy + z^3) \hat{i} + x^2 \hat{j} + 3xz^2 \hat{k}$ is conservative, and find the corresponding potential function. 10
- 4.(c) (i) Derive the relativistic law of addition of velocities from the Lorentz transformation. Two particles move directly towards each other, each with a speed $0.8c$ as measured in the laboratory; find the speed of one particle as measured by the other. 10
- (ii) The mean lifetime of a charged pion in its own rest frame is 2.6×10^{-8} s. If a beam of such pions moves through the laboratory at a speed of $0.95c$, calculate (I) the mean lifetime measured in the laboratory and (II) the mean distance travelled by a pion before decaying. 10
- 5.(a) For the scattering of a charged particle by a fixed Coulomb centre, derive the relation between the impact parameter b and the scattering angle θ . Also obtain an expression for the Rutherford scattering cross-section and discuss its significance. 20
- 5.(b) Muons are produced at a height of 3.0 km above the Earth's surface and travel vertically downward at a speed of $0.98c$. The mean lifetime of a muon in its own rest frame is 2.2×10^{-6} s. Find the mean distance the muons would travel before decaying if there were no time dilation. Taking time dilation into account, find the mean lifetime and the mean distance travelled in the Earth's frame, and comment on whether the muons can reach the ground. 15
- 5.(c) (i) Explain the phenomenon of precession of a spinning symmetric top under gravity, and derive an expression for the angular velocity of precession. 10
- (ii) A gyroscope wheel of moment of inertia 2.0×10^{-3} kg m² spins about its axis at 300 rad s⁻¹. Its mass is 0.5 kg and its centre of mass is 0.04 m from the pivot. Find the rate of precession. 5

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EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

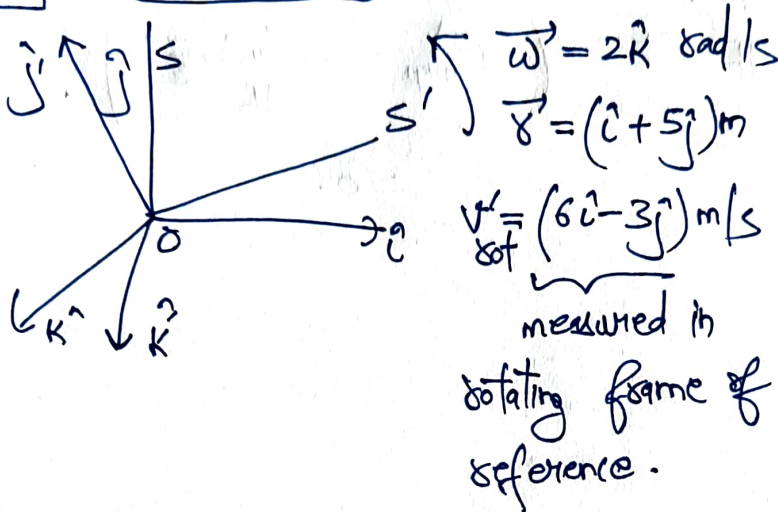
5.

6.

All the Best

1.(a)

A frame S' rotates with a constant angular velocity $\omega = 2 \hat{k} \text{ rad s}^{-1}$ relative to a fixed frame S having the same origin. At a certain instant a particle has position vector $\mathbf{r} = (1 + 5 \hat{j}) \text{ m}$ and velocity $\mathbf{v}' = (6 \hat{i} - 3 \hat{j}) \text{ m s}^{-1}$ as measured in S' . Determine the Coriolis acceleration and the centrifugal acceleration of the particle at this instant. 10



$$\vec{a}_c = -2(\vec{\omega} \times \vec{v}_{\text{rot}}) \quad (\text{Coriolis acceleration}).$$

$$= -2(2\hat{k} \times (2\hat{i} - \hat{j})) \times 3$$

$$\vec{a}_c = -12(\hat{k} \times (2\hat{i} - \hat{j})) \text{ m/s}^2 = -12(2\hat{j} + \hat{i}) \text{ m/s}^2$$

$$\boxed{\vec{a}_c = -12(2\hat{j} + \hat{i}) \text{ m/s}^2}$$

Centrifugal acceleration

$$\vec{a}_{\text{centri}} = -\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$= -2\hat{k} \times (2\hat{k} \times (1 + 5\hat{j})) \text{ m/s}^2$$

$$= -4\hat{k} \times [\hat{j} - 5\hat{i}]$$

$$= -4[-\hat{i} - 5\hat{j}] \text{ m/s}^2$$

$$\boxed{\vec{a}_{\text{centrifugal}} = 4(\hat{i} + 5\hat{j}) \text{ m/s}^2}$$

In this case, the Euler's acc'n will be

$$\text{as } \frac{d\vec{\omega}}{dt} = 0 \quad \left(\vec{a}_{\text{Euler}} = -\frac{d\vec{\omega}}{dt} \times \vec{r} \right) \\ = 0.$$

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1.(b)

A narrow tunnel is bored along a diameter of the Earth. Show that a body dropped into it executes simple harmonic motion, and find its time period.

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1.(b)

A narrow tunnel is bored along a diameter of the Earth. Show that a body dropped into it executes simple harmonic motion, and find its time period.

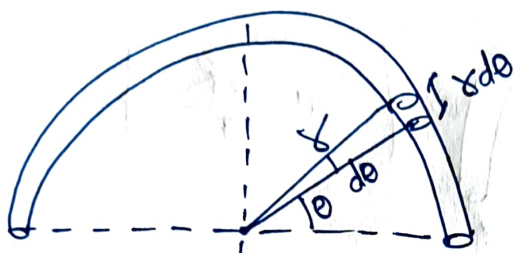
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1.(c)

Find the position of the centre of mass of a uniform semicircular wire (a half-ring) of radius R .

10

1(c).



Let λ = uniform
linear mass
density of
semicircular ring.

$x_{\text{com}} = 0$ due to symmetrical mass distribution about y axis.

$$\therefore y_{\text{com}} = \frac{\int dm y_p}{\int dm}$$

Consider a mass element $dm = \lambda(R d\theta)$.

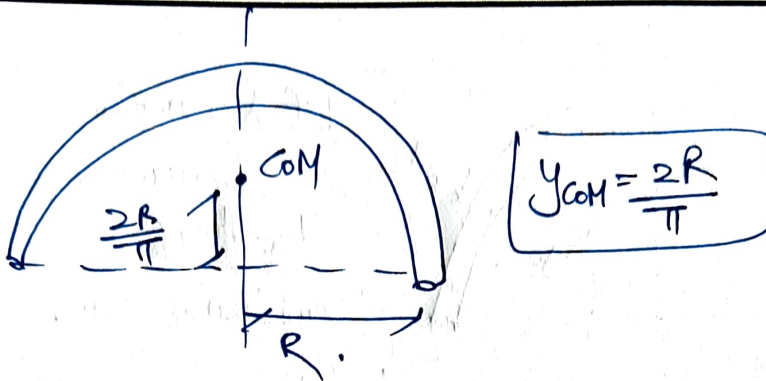
with $y_p = (R \sin \theta)$

$$M = \int dm = M \quad \& \quad \lambda = \frac{M}{\pi R}.$$

$$y_{\text{com}} = \frac{\int \lambda R d\theta \cdot R \sin \theta}{M} = \frac{\lambda R^2 \int_0^\pi \sin \theta d\theta}{M}$$

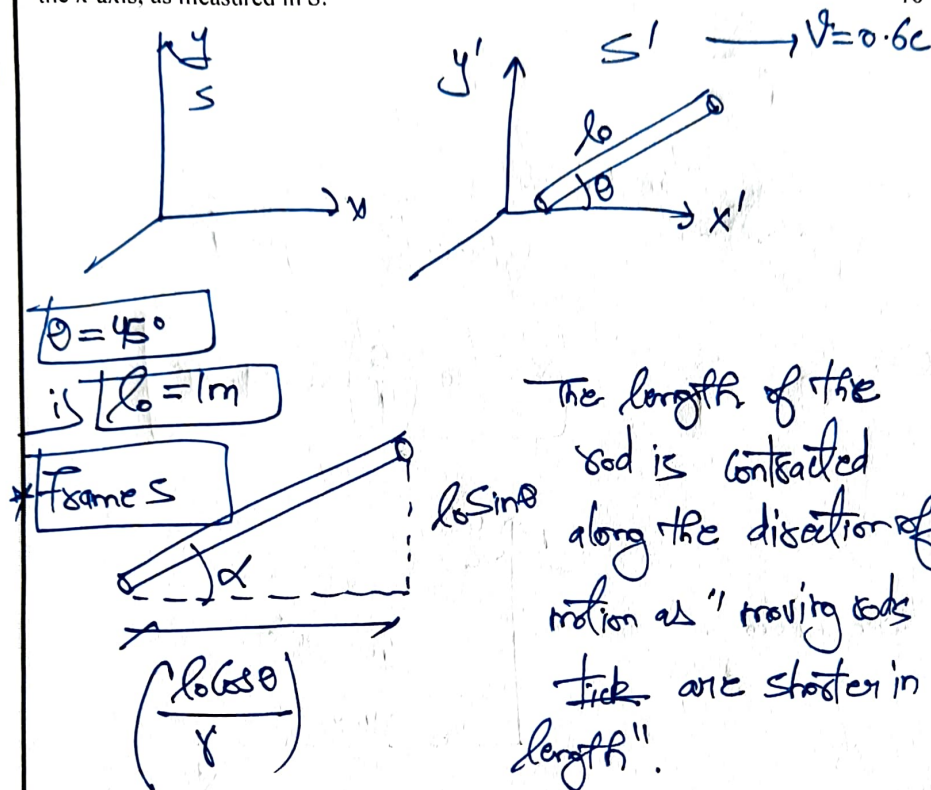
$$y_{\text{com}} = \frac{2\lambda R^2}{M} \quad \left(\because \int_0^\pi \sin \theta d\theta = 2 \right).$$

$$= \frac{2R^2}{M} \cdot \frac{M}{\pi R} \rightarrow \boxed{y_{\text{com}} = \frac{2R}{\pi}}$$



- 1.(d) A rod of proper length 1.0 m lies in the $x'y'$ -plane of its rest frame S' , making an angle of 45° with the x' -axis. The frame S' moves with a velocity $0.6c$ along the x -axis relative to the laboratory frame S . Determine (i) the length of the rod and (ii) the angle it makes with the x -axis, as measured in S . 10

1(d)



$$\therefore l^2 = l_0^2 \sin^2 \theta + \frac{l_0^2 \cos^2 \theta}{\gamma^2} = l_0^2 \left[\frac{\gamma^2 \sin^2 \theta + \cos^2 \theta}{\gamma^2} \right]$$

$$l^2 = l_0^2 \left[\sin^2 \theta + \left(1 - \frac{v^2}{c^2}\right) \cos^2 \theta \right]$$

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2} \cos^2 \theta} = (1) \sqrt{1 - (0.6)^2 \cos^2 45^\circ}$$

$$= 1 \sqrt{1 - \frac{0.36}{2}} = \sqrt{1 - 0.18}$$

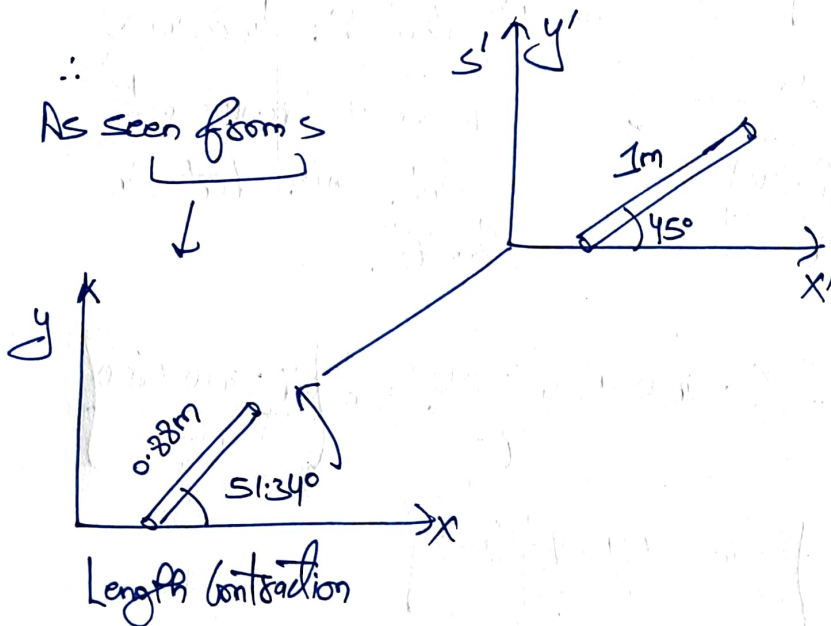
$$l = \sqrt{0.82} = 0.905\text{m}$$

$$\tan \alpha = \frac{L_0 \sin \theta}{\frac{L_0 \cos \theta}{\gamma}} = \gamma \tan \theta$$

$$\tan \alpha = \frac{1}{\sqrt{1-(0.5)^2}} \tan 45^\circ = \frac{1}{\sqrt{1-0.36}} = \frac{1}{\sqrt{0.64}}$$

$$\tan \alpha = \frac{10}{8} = \frac{5}{4}$$

$$\alpha = \tan^{-1}\left(\frac{5}{4}\right) = 51.34^\circ$$



1.(e)

An electron is accelerated from rest until it acquires a kinetic energy of 1.5 MeV.
Calculate its speed and its momentum.

10

1(e).

$$\text{Kinetic energy of } e^- = 1.5 \text{ MeV} \\ = 1.5 \times 10^6 \text{ eV.} \quad \text{--- (1)}$$

$$m_e = \text{mass of } e^- = 9.11 \times 10^{-31} \text{ kg.}$$

Kinetic energy in relativistic regime is

$$K = m_e c^2 (\gamma - 1), \quad \gamma > 1$$

$$\Rightarrow \gamma - 1 = \frac{K}{m_e c^2} \quad \propto \quad \boxed{\gamma = 1 + \frac{K}{m_e c^2}}$$

where $m_e c^2$ is the rest mass = 0.51 eV
energy of e^-

$$\therefore \gamma = 1 + \frac{1.5 \times 10^6 \text{ eV}}{0.51 \text{ eV}} \gg 1.$$

Since $K \gg m_e c^2$, $E = \frac{p^2}{2m}$

$$\therefore p = \sqrt{2mK} = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.5 \times 10^6 \times 1.602 \times 10^{-19}}$$

$$\boxed{p = 6.61 \times 10^{-22} \text{ kg m/s}^2}$$

2.(a)

A particle moves under the action of an attractive central force in a plane orbit described by $r = a(1 + \cos\theta)$, where a is a constant. Show that the magnitude of the force varies as the inverse fourth power of the distance from the centre of force. Also, calculate the potential energy and the total energy. 20

2(a)

For central force motion, writing Binet's eqⁿ

$$\frac{d^2u}{d\theta^2} + u = -\frac{mF\left(\frac{1}{u}\right)}{J^2u^2}, \quad u = \frac{1}{r}, \quad J = \text{Magnitude of Angular momentum of particle.}$$

Given $r = a(1 + \cos\theta)$

$$\therefore u = \frac{1}{a(1 + \cos\theta)} = \frac{1}{2a\cos^2\frac{\theta}{2}} = \frac{\sec^2\frac{\theta}{2}}{2a}$$

$$\frac{du}{d\theta} = \frac{1}{a} \frac{d}{d\theta} \left(\frac{1}{2\cos^2\frac{\theta}{2}} \right) = \frac{1}{2a} \left[\frac{d}{d\theta} \left(\sec^2\frac{\theta}{2} \right) \right]$$

$$= \frac{\sec^2\frac{\theta}{2}}{a} \cdot \sec^2\frac{\theta}{2} \cdot \tan\frac{\theta}{2} \cdot \left(\frac{1}{2}\right) = \frac{\sec^2\frac{\theta}{2} \tan\frac{\theta}{2}}{2a}$$

$$= \frac{1}{2a} \left[\tan\frac{\theta}{2} \left(1 + \tan^2\frac{\theta}{2} \right) \right]$$

$$\frac{d^2u}{d\theta^2} = \frac{1}{2a} \left[\sec^2\frac{\theta}{2} + 3\tan^2\frac{\theta}{2} \cdot \tan\frac{\theta}{2} \cdot \sec^2\frac{\theta}{2} \right] \cdot \frac{1}{2}$$

$$= \frac{1}{4a} \cdot \left(\sec^2\frac{\theta}{2} \right) \left[1 + 3\tan^2\frac{\theta}{2} \right]$$

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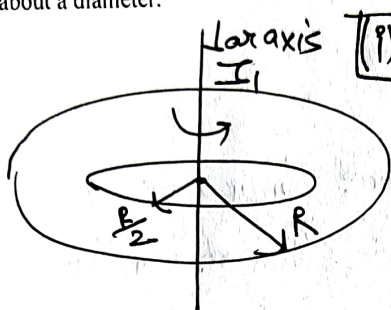
2.(b)

A uniform circular disc of mass M and radius R has a concentric circular hole of radius $R/2$ cut out of it. Determine the moment of inertia of the remaining body

- (i) about an axis through its centre perpendicular to its plane, and
(ii) about a diameter.

15

2(b)



(i) Moment of Inertia (MoI) about axis perpendicular to its plane = I_1

To calculate I_1 , assume the hollow or circular hole of radius $\frac{R}{2}$ to be filled with a negative surface mass density $(-\sigma)$ & a positive surface mass density $+\sigma$. This is to exploit the geometry for a clean calculation.

Since the MoI of a uniform disc $(mR^2) = \frac{mR^2}{2}$

$$\therefore I_1 = \frac{M_1 R^2}{2} - \frac{M_2 \left(\frac{R}{2}\right)^2}{2} = \frac{M_1 R^2}{2} - \frac{M_2 R^2}{8}$$

$$M = \text{Area} \times \sigma \quad \therefore M_1 = \pi R^2 \sigma$$

$$M_2 = -\pi \left(\frac{R}{2}\right)^2 \sigma$$

$$\sigma = \frac{M}{\pi R^2}$$

$$\therefore I_1 = \frac{\pi \sigma R^4}{2} - \frac{\pi R^4 \sigma}{32} = \frac{\pi R^4 \sigma}{2} \left(1 - \frac{1}{16}\right)$$

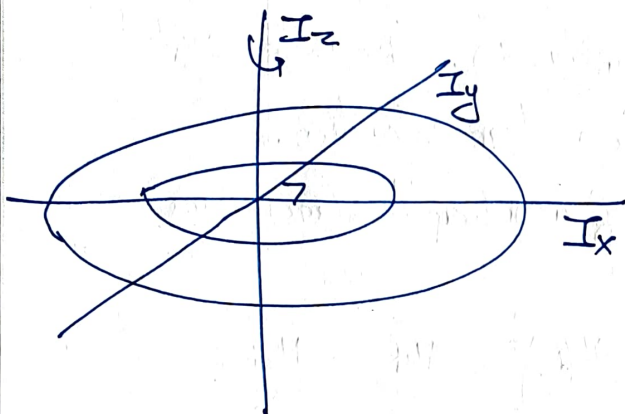
$$I_1 = \frac{15}{32} \pi R^4 \sigma$$

$$I = \frac{15}{32} \pi R^4 \left(\frac{M}{\pi R^2} \right)$$

$$I = \frac{15MR^2}{32}$$

(ii) Since the hollow disc is a plane lamina or 2D rigid body, Perpendicular axis theorem can be applied,

Let $I_z = \frac{15MR^2}{32}$ is along the z-axis vertical.



Consider two perpendicular axis in the plane of disc, intersecting at the centre of disc, (be I_x & I_y).

$$\therefore I_z = I_x + I_y$$

Due to symmetry, $I_x = I_y \Rightarrow$

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$$2I = I_2$$

$$\therefore I = \frac{I_2}{2}$$

$$I = \frac{15}{64} MR^2$$

- 2.(c) Starting from the expression for relativistic momentum $p = \gamma m_0 v$, use the work energy theorem to derive the relativistic kinetic energy $K = (\gamma - 1)m_0 c^2$. Hence show that for $v \ll c$ it reduces to $\frac{1}{2}m_0 v^2$, and that for a particle of total energy E and momentum p the speed is $v = pc^2/E$. 15

2(c). Relativistic momentum $p = \gamma m_0 v$

Work energy theorem, $W_{net} = \Delta K$

$$\therefore K_b - K_a = \int \vec{F} \cdot d\vec{x}$$

$\vec{F} = \frac{d\vec{p}}{dt}$, substituting in above eqⁿ,

$$K_b - K_a = \int_{x_a}^{x_b} \frac{d\vec{p}}{dt} \cdot d\vec{x} = \int_{x_a}^{x_b} v dp$$

$p = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$, so substituting in above eqⁿ,
and applying integration by parts,

$$K_b - K_a = \int_{x_a}^{x_b} v \underbrace{d\left(\frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}\right)}_{(2)}$$

$$= \frac{v^2 m_0}{\sqrt{1 - \frac{v^2}{c^2}}} - \int_{x_a}^{x_b} \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}} dv$$

$$1 - \frac{v^2}{c^2} = t \quad \therefore \quad -\frac{2v dv}{c^2} = dt \quad \text{or} \quad \frac{-v dv}{c^2} = \frac{dt}{2}$$

$$\Delta + m_0 \int \frac{dt}{2\sqrt{t}} = m_0 c^2 \sqrt{t}$$

$$K_b - K_a = \frac{m_0 v^2}{\sqrt{1 - \frac{v^2}{c^2}}} \bigg|_{v_a}^{v_b} + c^2 m_0 \sqrt{1 - \frac{v^2}{c^2}} \bigg|_{v_a}^{v_b}$$

choosing initial conditions as $K_a = 0$, $K_b = K$
 $v_a = 0$, $v_b = v$

$$\therefore K = \frac{m_0 v^2}{\sqrt{1 - \frac{v^2}{c^2}}} + m_0 \left[\sqrt{1 - \frac{v^2}{c^2}} - 1 \right] c^2$$

$$K = m_0 c^2 \left[\frac{\left(\frac{v^2}{c^2}\right)}{\sqrt{1 - \frac{v^2}{c^2}}} \right] + m_0 \left[\sqrt{1 - \frac{v^2}{c^2}} - 1 \right] c^2$$

$$K = m_0 c^2 \gamma \left[\cancel{\frac{v^2}{c^2}} + \left(1 - \cancel{\frac{v^2}{c^2}}\right) - \sqrt{1 - \frac{v^2}{c^2}} \right]$$

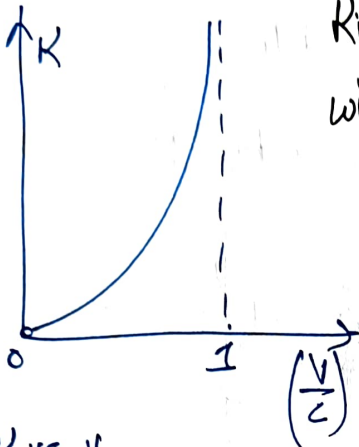
$$K = m_0 c^2 \gamma \left[1 - \frac{1}{\gamma} \right] \Rightarrow \boxed{K = m_0 c^2 (\gamma - 1)}$$

For $v \ll c$, the above formula shows resemblance with its classical counterpart i.e. $K = \frac{1}{2} m_0 v^2$

$$K = m_0 c^2 \left[\left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} - 1 \right] \quad \text{for } v \ll c \quad \text{(Binomial approx.)}$$

$$K = m_0 c^2 \left[1 + \frac{v^2}{2c^2} - 1 \right] = \boxed{\frac{1}{2} m_0 v^2}$$

$$K = m_0 c^2 (\gamma - 1)$$



when v approaches c ,
Kinetic energy increases
without bounds $\Rightarrow$ No
massive particle can
attain the speed of light

K vs $\frac{v}{c}$

From Mass energy
invariance,

$$E^2 = p^2 c^2 + m_0^2 c^4$$

differentiating with respect to p ,

$$2E \frac{dE}{dp} = 2pc^2 \Rightarrow \boxed{\frac{dE}{dp} = \frac{pc^2}{E}}$$

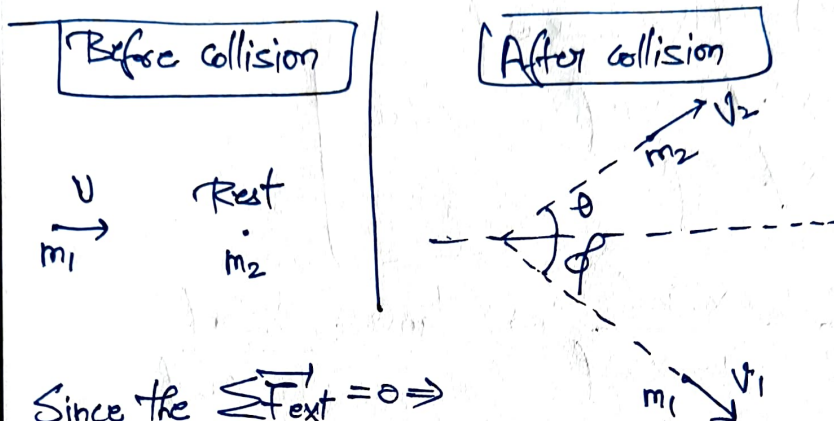
$$\boxed{\frac{dE}{dp} = \frac{\gamma m_0 v c^2}{\gamma m_0 c^2} = v}$$

3.(a)

(i) A particle of mass m_1 moving with velocity u collides head-on and elastically with a particle of mass m_2 initially at rest. Derive expressions for the velocities of the two particles after the collision.

10

3(a)



Since the $\sum \vec{F}_{ext} = 0 \Rightarrow$

Linear momentum is conserved,

$$\vec{P}_i = \vec{P}_f \Rightarrow m_1 u = m_1 v_1 \cos \phi + m_2 v_2 \cos \theta$$

$$\& \quad m_2 v_2 \sin \theta = m_1 v_1 \sin \phi$$

Now, $\frac{v_1}{u} = x, \quad \frac{v_2}{u} = y \quad \therefore \text{Let } x \neq \frac{m_1}{m_2}$

$$m_1 = m_1 x \cos \phi + m_2 y \cos \theta$$

$$x = \frac{m_2}{m_1}$$

$$\therefore x \cos \phi + y x \cos \theta = 1 \quad \text{--- (1)}$$

$$y x \sin \theta = x \sin \phi \quad \text{--- (2)}$$

$$E_i = E_f \quad \therefore \frac{P_i^2}{2m_1} = \frac{P_1'^2}{2m_1} + \frac{P_2'^2}{2m_2}$$

$$\therefore \frac{m_1 u^2}{2} = \frac{m_1 v_1^2}{2} + \frac{m_2 v_2^2}{2} \Rightarrow 1 = x^2 + y^2 \quad \text{--- (3)}$$

$$\& \quad \text{(1)}^2 - \text{(2)}^2 \Rightarrow y^2 x^2 = x^2 \sin^2 \phi + (1 - x \cos \phi)^2$$

$$y^2 x^2 = 1 + x^2 - 2x \cos \phi \rightarrow x - x x^2 = 1 + x^2 - 2x \cos \phi$$

$$x(1 - x^2) = 1 + x^2 - 2x \cos \phi$$

$$\begin{array}{l} x \\ x \end{array} \left[\begin{array}{l} 2x^2 \neq 2x \cos \phi \\ x \neq \cos \phi \end{array} \right] \quad x \neq 0$$

$$\frac{xy \sin \theta}{xy \cos \theta} = \frac{x \sin \phi}{1 - x \cos \phi} \Rightarrow \tan \theta = \frac{x \sin \phi}{1 - x \cos \phi}$$

$$2x^2(1 + x) - (2 \cos \phi)x + (1 - x) = 0$$

$$x = \frac{2 \cos \phi \pm \sqrt{4 \cos^2 \phi - 2(1 - x^2)}}{2(1 + x)} = \frac{V_1}{U} \quad \begin{array}{l} V_1 \\ \text{in terms} \\ \text{of} \\ U, \phi. \end{array}$$

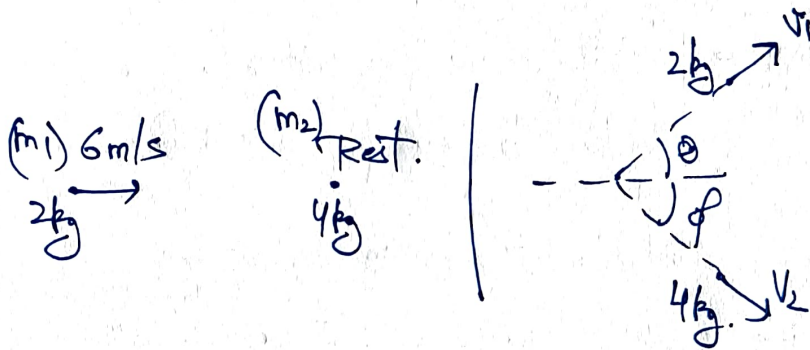
$$= 2 \cos \phi \pm \underline{\hspace{2cm}}$$

(ii) A body of mass 2 kg moving at 6 m s⁻¹ collides head-on and elastically with a stationary body of mass 4 kg. Find the velocities of both bodies after the collision. 5

Kinetic energy remains conserved in CoM mass frame,

$$V_{\text{CoM}} = \frac{(2 \times 6)}{2+4} = 2 \text{ m/s}$$

$$\therefore \frac{1}{2} M_1 V_{1c}^2 + \frac{1}{2} M_2 V_{2c}^2$$



$$\tan \theta = \frac{\sin 2\phi}{1 - \cos 2\phi}, \quad \gamma = \frac{m_1}{m_2}$$

$$\tan \theta = \frac{\sin 2\phi (2)}{1 - 2 \cos 2\phi}$$

3.(b)

(i) State Bernoulli's theorem and derive it for the streamline flow of an ideal, incompressible, non-viscous fluid. 10

Bernoulli's theorem - it is essentially a restatement of conservation of Energy. The total energy of a fluid element flowing along a streamline in an incompressible & inviscid fluid remains constant, i.e. $P + \frac{1}{2} \rho v^2 + \rho gh = \text{const.}$

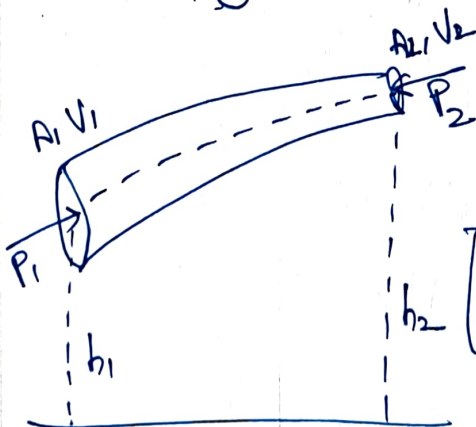
where P = Pressure energy per unit volume

$\frac{1}{2} \rho v^2$ = Kinetic energy per unit vol.

ρgh = Potential energy per unit volume.

Consider a fluid element flowing across a streamline from section 1 to section 2 as shown in figure.

where A = Area of cross section



v = Speed along streamline.

Work done by Pressure driven force

$$W_1 = P_1 A_1 v_1 \Delta t$$

$$W_1 = P_1 \Delta V$$

$$W_2 = -P_2 A_2 v_2 \Delta t$$

Since $A_1 V_1 = A_2 V_2$ (Equation of continuity for incompressible fluids).

$$\therefore \underbrace{A_1 V_1 \Delta t}_{V_1} = \underbrace{A_2 V_2 \Delta t}_{V_2 \text{ (volume)}} = \Delta V \text{ (Volume of fluid flowing through)}$$

$$\therefore \boxed{W_{\text{Pressure}} = (P_1 - P_2) \Delta V}$$

Work done by ~~the~~ Gravity - Since gravity is a conservative force,

$$W_{\text{gravity}} = -\Delta U = -(U_2 - U_1)$$

$$W_{\text{gravity}} = U_1 - U_2$$

$$\begin{aligned} W_{\text{gravity}} &= m_1 g h_1 - m_2 g h_2 = \rho V_1 h_1 g - \rho V_2 h_2 g \\ &= \rho (A_1 V_1 \Delta t) h_1 g - \rho (A_2 V_2 \Delta t) h_2 g \end{aligned}$$

$$\boxed{W_{\text{gravity}} = \rho g (h_1 - h_2) \Delta V}$$

From work energy theorem, $W_{\text{net}} = \Delta K$

$$W_{\text{Pressure}} + W_{\text{gravity}} = K_2 - K_1 = \frac{1}{2} m_2 V_2^2 - \frac{1}{2} m_1 V_1^2$$

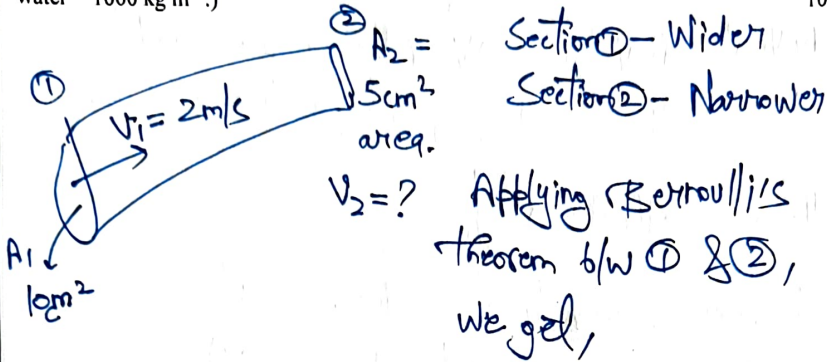
$$(P_1 - P_2) \Delta V + \rho g (h_1 - h_2) \Delta V = \frac{\Delta V}{2} (\rho V_2^2 - \rho V_1^2)$$

$$\therefore P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g h_2 = \text{Constant}$$

$$\boxed{P + \rho g h + \frac{1}{2} \rho V^2 = C}$$

Static Pressure + Dynamic pressure = Constant

(ii) Water flows steadily through a horizontal pipe whose cross-sectional area narrows from 10 cm^2 to 5 cm^2 . If the speed of water in the wider section is 2 m s^{-1} , find the speed in the narrower section and the pressure difference between the two sections. (Density of water = 1000 kg m^{-3} .)



$$P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g h_2$$

Since the pipe is horizontal, $\rho g h_1 = \rho g h_2$

$$\therefore P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2 \quad \text{--- ①}$$

Using equation of Continuity i.e. the amount of fluid that enters section ① must be equal to the fluid leaving section ②

$$\text{i.e. } A_1 V_1 = A_2 V_2$$

$$\frac{10^{-4} \text{ m}^2}{1 \text{ cm}^2} \times (10 \text{ cm}^2) (2 \text{ m/s}) = (5 \text{ cm}^2) V_2 \times \frac{10^{-4} \text{ m}^2}{1 \text{ cm}^2}$$

$$\therefore \boxed{V_2 = 4 \text{ m/s}}$$

$$\text{Given } \rho = 1000 \text{ kg/m}^3.$$

$$(P_1 - P_2) = \text{Pressure difference across pipe}$$

$$= \frac{1}{2} \rho (V_2^2 - V_1^2) = \frac{1}{2} \times 1000 \times (16 - 4) = 6000 \text{ N/m}^2$$

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$$\therefore \Delta P = 6000 \text{ N/m}^2$$

3.(c)

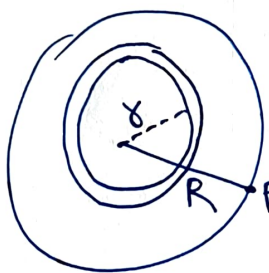
A solid sphere of radius R has a density that varies with the distance r from its centre as $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant. Obtain expressions for the gravitational field due to the sphere at its surface and at a point distant $R/2$ from the centre.

15

3(c) Varying mass density $= \rho_0(1 - \frac{r}{R})$

$\rho_0 = \text{Constant.}$

Gravitational field at its surface - will be due to all the mass enclosed within $r \leq R$.



Consider a thin spherical shell of mass (dm) & thickness dr at a distance r from the centre of solid sphere.

The Gravitational field due to this shell at point P (at $r=R$) will be $dE = -\frac{G dm}{R^2}$

$$dE = -\frac{G}{R^2} \cdot \rho(r) dV = -\frac{G}{R^2} \cdot \rho_0 \left(1 - \frac{r}{R}\right) (4\pi r^2 dr)$$

$$E = -\frac{4\pi G \rho_0}{R^2} \int_0^R \left(1 - \frac{r}{R}\right) r^2 dr \quad \left[\text{Net field at point P due to all such infinitesimally thin shells} \right]$$

$$E(r=R) = -\frac{4\pi G \rho_0}{R^2} \int_0^R \left(r^2 - \frac{r^3}{R} \right) dr$$

$$= -\frac{4\pi G \rho_0}{R^2} \cdot \left[\frac{r^3}{3} - \frac{r^4}{4R} \right]_0^R$$

$$= -\frac{4\pi G \rho_0}{R^2} \left[\frac{R^3}{3} - \frac{R^3}{4} \right]$$

$$= -4\pi G \rho_0 R \left[\frac{1}{3} - \frac{1}{4} \right]$$

$$\boxed{\vec{E}(r=R) = \frac{-\pi G \rho_0 R}{3} \hat{r}}$$

where

$\hat{r}$ = unit vector from centre to P.

$$\boxed{\text{Gravitational field at } r = \frac{R}{2}}$$

From Gauss's law, the field at $r = \frac{R}{2}$ will be ONLY due to the mass enclosed within $r \leq \frac{R}{2}$.

$$\therefore E = \int dE = -\frac{4\pi G \rho_0}{R^2} \int_0^{\frac{R}{2}} \left(1 - \frac{r}{R} \right) r^2 dr$$

$$E = -\frac{4\pi G \rho_0}{R^2} \left[\frac{r^3}{3} - \frac{r^4}{4R} \right]_0^{\frac{R}{2}} =$$

$$E = \frac{-4\pi G \rho_0}{R^2} \left[\frac{R^3}{24} - \frac{R^4}{64R} \right]$$

$$= -\pi G \rho_0 R \left[\frac{1}{6} - \frac{1}{16} \right]$$

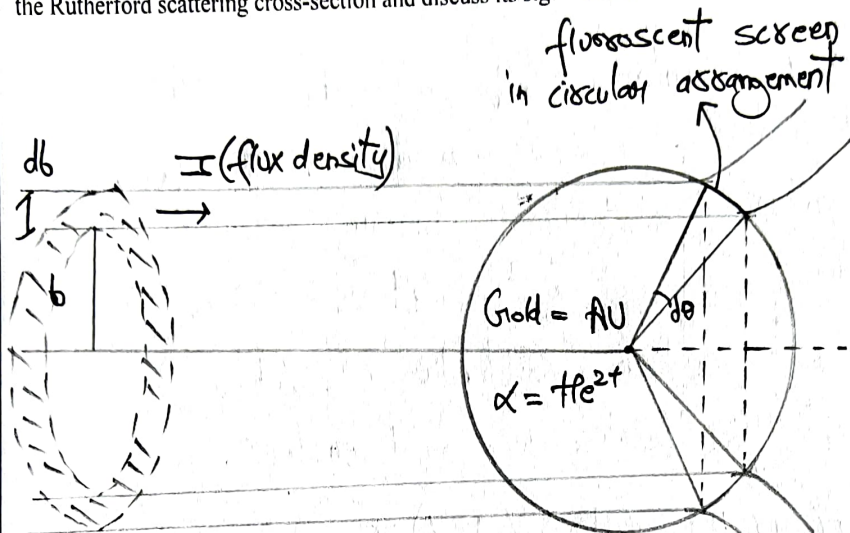
$$= \frac{-\pi G \rho_0 R}{2} \left(\frac{8-3}{24} \right) = -\pi G \rho_0 \left(\frac{5}{48} \right) R$$

$$\therefore \boxed{\vec{E} \left(r = \frac{R}{2} \right) = -\frac{5}{48} \pi G \rho_0 R}$$

5.(a)

For the scattering of a charged particle by a fixed Coulomb centre, derive the relation between the impact parameter b and the scattering angle θ . Also obtain an expression for the Rutherford scattering cross-section and discuss its significance. 20

5(a)



Setup: Rutherford Scattering experiment.

In this classical experiment, a large number of Alpha particles (doubly charged Helium nucleus) were bombarded on a very thin gold foil, a fluorescent screen placed in a circular arrangement detected the scattered α particles in the form of Flashes of light.

Thin Gold foil was taken so that the α -ray gets scattered by only 1 nucleus.

Consider vars $I = \text{Flux density} = \text{No. of } \alpha \text{ particle passing normal to incident beam per unit area per unit time}$

$b = \text{Impact parameter.}$

$\frac{d\sigma}{d\Omega}$ = differential scattering cross section.

Consider a circular strip of thickness " db " around radius = impact parameter. Therefore, the number of α particles passing through $(2\pi b db)$ area must get scattered through the fluorescent screen subtending a solid angle $d\Omega'$ at the position of Gold nucleus.

$$\therefore I(2\pi b db) = \left(\frac{d\sigma}{d\Omega}\right) (d\Omega') I$$

where $d\Omega' = \text{Solid angle} = \frac{\text{area of circular strip}}{r^2}$

$$= \frac{2\pi r \sin\theta dr}{r^2} = 2\pi \sin\theta d\theta$$

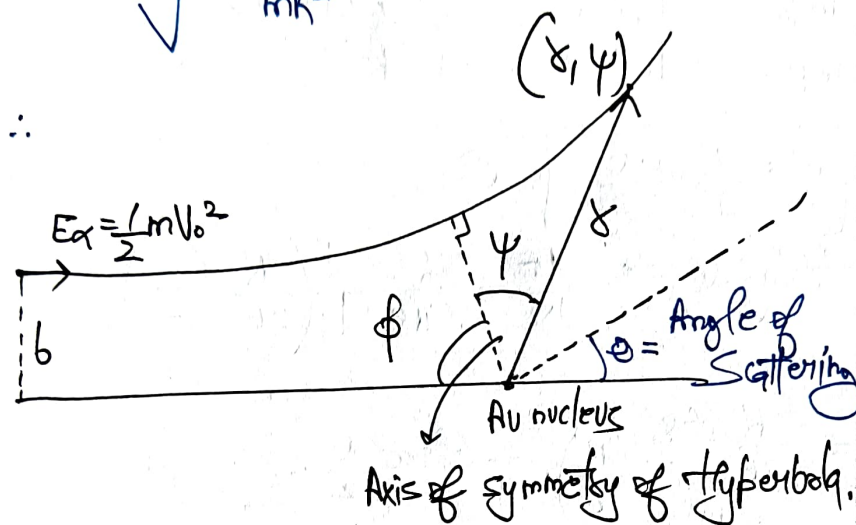
$$\therefore I(2\pi b db) = \left(\frac{d\sigma}{d\Omega}\right) (2\pi \sin\theta d\theta) I$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|} \quad - (1)$$

Calculating ' b ' as a function of θ (θ). Since the Coulombic force is central in nature, the trajectory will be $\frac{1}{r} = 1 + e \cos\psi$

Since this is not a bound system $\Rightarrow E > 0$

$$\therefore e = \sqrt{1 + \frac{2EJ^2}{mK^2}} > 1 \Rightarrow \text{hyperbola.}$$



As $r \rightarrow \infty$, $\psi \rightarrow \phi$ $\therefore$ in $\frac{r}{b} = 1 + e \cos \psi$ becomes

$$0 = 1 + e \cos \phi \Rightarrow \cos \phi = -\frac{1}{e}$$

$$\text{Now, } 2\phi + \theta = \pi \Rightarrow \phi = \frac{\pi - \theta}{2}$$

$$\therefore \sin \frac{\theta}{2} = \frac{1}{e} \propto -e = \sec \frac{\theta}{2}$$

$$e^2 = \sec^2 \frac{\theta}{2} \therefore 1 + \frac{2EJ^2}{mK^2} = \sec^2 \frac{\theta}{2}$$

J = Angular momentum of α particle, which remains conserved because $\vec{F} \parallel \vec{r}$ $\therefore$

$$\boxed{J = m v_0 b}$$
, Substituting in above equation,

$$\frac{2E(mv_0b)^2}{mK^2} = \frac{K^2 \theta}{2}$$

$$\frac{2Emv_0^2 b^2}{mK^2} = \frac{K^2 \theta}{2} \quad \text{or} \quad \left(\frac{2Eb}{K}\right)^2 = \frac{K^2 \theta}{2}$$

$$\Rightarrow \boxed{b = \frac{K}{2E} \frac{K^2 \theta}{2}} \quad \text{--- (2)}$$

Substituting in (1) $\Rightarrow \frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta} \frac{d}{d\theta} \left(\frac{K}{2E} \frac{K^2 \theta}{2} \right)$

$$\frac{d\sigma}{d\Omega} = \frac{\frac{K}{2E} \frac{K^2 \theta}{2}}{\sin\theta} \cdot \left(\frac{K}{2E} \right) \cdot -\sec^2 \theta \cdot \frac{1}{2}$$

$$= \left(\frac{K}{2E} \right)^2 \frac{\cancel{\frac{K^2 \theta}{2}}}{\sin\theta} \cdot \frac{1}{2\sin\theta} \cdot \frac{\cancel{\frac{K^2 \theta}{2}}}{2} \cdot \frac{1}{2}$$

$$= \left(\frac{K}{4E} \right)^2 \frac{1}{\sin^4 \frac{\theta}{2}} \quad \text{or} \quad \boxed{\frac{d\sigma}{d\Omega} = \left(\frac{K}{4E} \right)^2 \sec^4 \frac{\theta}{2}}$$

Significance ① $\rightarrow$ For more energetic α rays, the scattering through θ is less.

② Inversely proportional to $\sin^4 \frac{\theta}{2}$.

③ For high energy α rays on a light nucleus (less mass number), the α rays can reach very close to nucleus. $\Rightarrow R = R_0 A^{\frac{1}{3}}$, $R_0 = 1.4 \text{ fm}$, $A = \text{mass No.}$

5.(b)

Muons are produced at a height of 3.0 km above the Earth's surface and travel vertically downward at a speed of 0.98c. The mean lifetime of a muon in its own rest frame is 2.2×10^{-6} s. Find the mean distance the muons would travel before decaying if there were no time dilation. Taking time dilation into account, find the mean lifetime and the mean distance travelled in the Earth's frame, and comment on whether the muons can reach the ground.

15

5(b).

$$\text{Time measured in Muon's rest frame} = \text{Proper time } (\tau) = 2.2 \times 10^{-6} \text{ s}$$

Muon travelling with relativistic speed = 0.98c.

Mean distance travelled by Muons (No time dilation).

$$d = v\tau = (0.98c)(\tau) = (0.98c \times 2.2 \times 10^{-6}) \text{ m}$$

$$\boxed{d = 646.8 \text{ m}}$$

But for an observer on earth, the time taken by muon will be more than τ as he will say "Moving clocks tick slow".

Moving clock - clock measuring the proper time (τ) of Muon.

$\therefore$ The consequence of Lorentz transformation, $t = \gamma\tau$ is the time measured in Earth's rest frame.

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - (0.98)^2}} = 5.03 \therefore \boxed{\gamma = 5.03}$$

Mean lifetime of Muon in Earth's rest frame

$$\tau_{\text{frame}} = \gamma \tau = (5.03) \times 2.2 \mu\text{s}$$

$$\boxed{t = 11.06 \mu\text{s}}$$

Mean distance travelled in earth's rest frame

$$\begin{aligned} \tau_{\text{frame}} &= vt \\ &= (0.98c)(11.06 \mu\text{s}) \\ &= 0.98 \times 10^8 \times 3 \times 11.06 \times 10^{-6} \text{ m} \\ &= 3251.64 \text{ m} \\ \text{or } \boxed{d = 3.25 \text{ km}} \end{aligned}$$

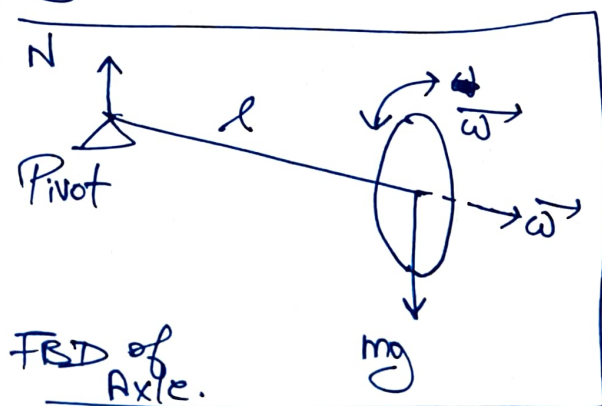
That's why the muons are detected by sensors & detectors placed in oceans at $d > 3 \text{ km}$ (where the muons are produced).

5.(c)

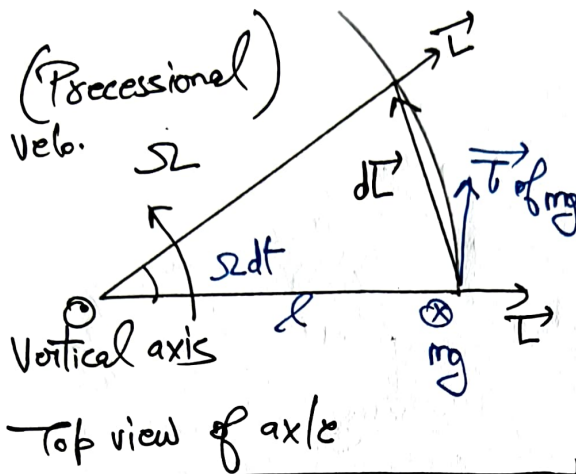
(i) Explain the phenomenon of precession of a spinning symmetric top under gravity, and derive an expression for the angular velocity of precession.

10

Consider a wheel given an initial spin angular velocity about its axle which has a pivot at 'O'. This pivot doesn't constrain the wheel except for providing a normal reaction to balance its weight. The rotating wheel, instead of falling down like a pendulum, starts wobbling for a while & enters into precession -al motion with angular velocity Ω about the vertical axis. This is the phenomenon of precession. There can be many formulations of the same & hence many applications of Gyroscope as well.



$N = mg$ (Vertical equilibrium)



Since the L is changing, there has to be a corresponding Torque responsible for it. For dt time, The L magnitude

remains same $\therefore L(\Omega dt) = dL$

$$\therefore \tau = L\Omega$$

Also, $\tau_{mg} = mgl (\text{dis'n})$.

$$\therefore mgl = L\Omega \Rightarrow \boxed{\Omega = \frac{mgl}{I\omega}}$$

where $I = \text{Moment of Inertia about Axle.}$

$\omega = \text{Angular velocity about axle.}$

Applications

- ① VR headsets, mobiles use Gyroscopic sensors.
- ② Used in ship stabilization.
- ③ Navigation in Aircrafts.
- ④ Bullets are given some initial spin so as to resist the deflective forces (mg).

(ii) A gyroscope wheel of moment of inertia $2.0 \times 10^{-3} \text{ kg m}^2$ spins about its axis at 300 rad s^{-1} . Its mass is 0.5 kg and its centre of mass is 0.04 m from the pivot. Find the rate of precession.

5

$$\omega = 300 \text{ rad/s}$$

$$m = 0.5 \text{ kg}$$

$$\Omega = \frac{mgl}{I_0 \omega}$$

$$l = 0.04 \text{ m}$$

$$I_0$$

$$\therefore \Omega = \frac{mgl}{I_0 \omega} = \frac{(0.5)(9.8)(0.04)}{2 \times 10^{-3} \times 300} = 0.326 \text{ rad/s}$$

$$\therefore \boxed{\Omega = 0.326 \text{ rad/s}}$$

Rate of precession, independent of θ .

